

Core java

Intro

```
public class App {
    public static void main(String[] args) throws Exception {
        System.out.println("Hello, World!");

        // java=> high level language -> human understandable code
        // java => 1995 -> james gosling

        // java => hybrid model
        // compiler =>
        // interpreter =>

        // WORA => write once run anywhere

        // java => platform independent but jdk is dependent

        // app.java => app.class

        // platform => windows, macos, linux

        // jdk => java development kit
        //     JRE => java runtime environment (provides libraries )
        //     JVM => java virtual machine (interpreter software)

        // java => object oriented programming -> encapsulation,
abstraction , polymorphism, inheritance
        // javascript => object-basis language

        // object oriented => principle

        // java=> web application -> spring boot
        // webapp -> eg.flipkart
        // website -> eg.blog website

        // let data = 10 ;

        // int data =10;
        // data ="harini";

        // let data =10;
        // data = "harini"

        // java=> static typed language
        // javascript => dynamically typed language
    }
}
```

```

        // variables => identifier
        int data = 10;
        System.out.println(data);
        // ram => volatile memory ->
        // rom => non-volatile memory ->
        // data types
    }
}

```

1. What is Java?

- **Java** is a high-level, object-oriented programming language created in **1995 by James Gosling** at Sun Microsystems (now owned by Oracle).
- It follows the principle of **WORA (Write Once, Run Anywhere)** → compiled code runs on any platform with a Java Virtual Machine (JVM).

2. Java Architecture

- **JDK (Java Development Kit)** → tools for development (compiler, debugger).
- **JRE (Java Runtime Environment)** → libraries + JVM to run applications.
- **JVM (Java Virtual Machine)** → interpreter that executes bytecode.
- Flow:

Code

Source Code (.java) → Compiler → Bytecode (.class) → JVM → Execution

3. Features of Java

- **Platform Independent** → bytecode runs on any OS with JVM.
- **Object-Oriented** → supports encapsulation, inheritance, polymorphism, abstraction.
- **Robust** → strong memory management, exception handling.
- **Secure** → no explicit pointers, runs inside JVM sandbox.
- **Multithreaded** → supports concurrent execution.
- **Distributed** → supports networking and RMI.

4. Java vs JavaScript

- **Java** → statically typed, compiled + interpreted, used for enterprise apps, Android, backend (Spring Boot).
- **JavaScript** → dynamically typed, interpreted, used for web front-end, Node.js backend.

5. Applications of Java

- **Web Applications** → Spring Boot, JSP, Servlets.
- **Mobile Applications** → Android development.
- **Enterprise Applications** → Banking, ERP systems.
- **Desktop Applications** → Swing, JavaFX.
- **Big Data & Cloud** → Hadoop ecosystem, cloud-native apps.

Practice Tasks

1. Write a simple Java program that prints your name and age.
2. Demonstrate variable declaration with different data types (int, double, String, boolean).
3. Show the difference between `System.out.print()` and `System.out.println()`.
4. Create a program that calculates the sum of two numbers.
5. Write a program that demonstrates type casting (int → double, double → int).